

CI: $\rightarrow PM$

MT: Given PM $\rightarrow$ and we have verify its correctness

Ex: Apollo Tyres $\rightarrow$ Cars Tyres

They claim that their new tyres last
4500 kms.
PM

① State the Hypotheses

Null Hypothesis (H_0 or H_{10}): Default assumption
about any scenario

NH contains: $\geq, \leq, =$

Alternative Hypothesis (H_1 or H_A): Contradictory
assumption to Null Hypothesis or something
we hope to prove.

AH contains: $>, <, \neq$

② Objective: Try to reject the Null Hypothesis

③ How are we going to reject NH?

We reject it if we have significantly strong evidence against it.

④ What will be evidence for us, how does it look like?

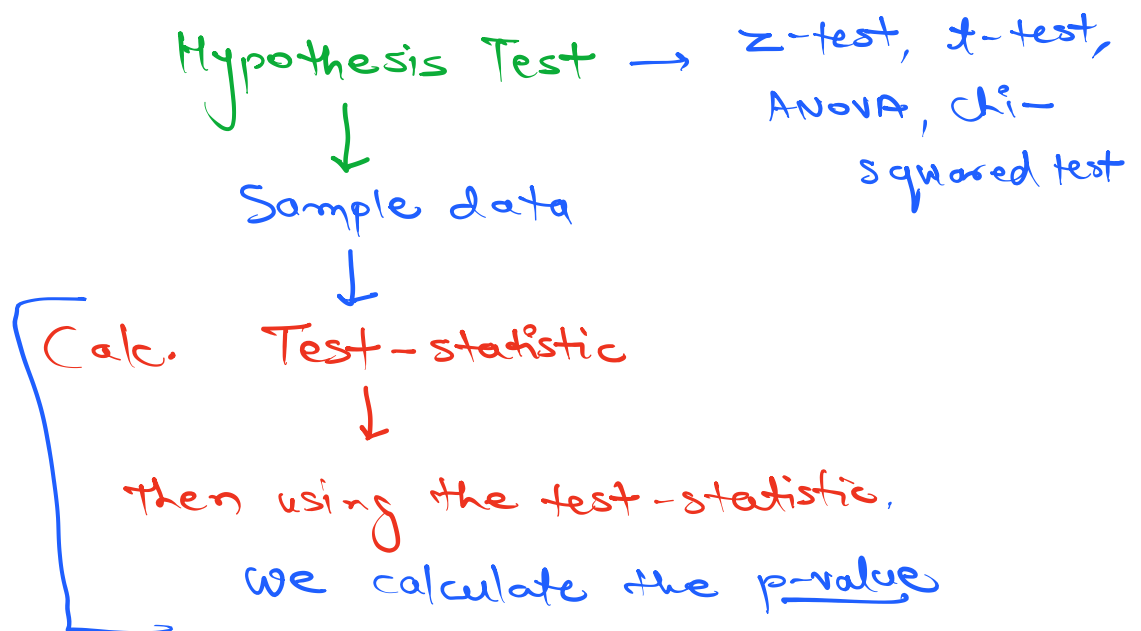
Evidence for us will be a value which is called "p-value".

⑤ To calculate the p-value,

we need to calculate 'Test-statistic'

first and using the test-statistic value we get the p-value

⑥ How will we get the evidence (p-value)?



⑦

Z-test
↓
Sample Data
↓
Z-statistic

$$\text{Z-stat} = \frac{SM - PM}{PSD / \sqrt{SS}}$$

= Number
↓
p-value

ANOVA
↓
SD
↓
F-statistic
↓
p-value

t-test
↓
Sample data
↓
t-statistic

$$t\text{-stat} = \frac{SM - PM}{SSD / \sqrt{SS}}$$

= Number
↓
p-value

Chi-squared test
↓
SD
↓
Chi-statistic
↓
p-value